Integrated Operation Enhancement: Resolving Processing Delays and Improving Management

A Proposal report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project titled " **Integrated Operation Enhancement: Resolving Processing Delays and Improving Management**." I extend my appreciation to **Balaji Powder Coating** for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability. Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report. I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. If plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Signature of Candidate:

Manami Bera

Name: Manami Bera

Date:08/10/2025

Executive Summary

Balaji Powder Coating, led by Mr. Arup Bera, is a specialized industrial powder-coating facility situated in Lakshmanpur Industrial Estate, Tentul Kuly, Domjur, Howrah, West Bengal, 711409, India. The company provides advanced surface finishing solutions for a wide range of metal products, including aluminium frames, battery boxes, beams, metal racks, steel furniture, almirahs, panel boxes, and driver desks. Serving clients across electrical, fabrication, and construction industries, Balaji Powder Coating plays a key role in enhancing product durability, corrosion resistance, and aesthetic quality. The factory's operations involve surface preparation, powder application, curing, and quality inspection — forming the core of its precision-driven manufacturing process.

Despite its technical expertise, the organization faces notable challenges in managing process flow and inventory levels. Unstructured order handling, manual data recording, and delayed coordination between production and dispatch often result in processing delays and material mismatches. These issues hinder efficient scheduling, increase operational costs, and effect on-time delivery rates.

This proposal aims to strengthen Balaji Powder Coating's operations through an integrated, data-driven enhancement framework. The approach includes systematic demand forecasting, workflow time analysis, and digital inventory tracking to optimize resource utilization. Implementing these measures will help reduce delays, improve productivity, and enable better forecasting accuracy—positioning the company for sustained efficiency and growth under the leadership of Mr. Arup Bera.

Organization Background

Balaji Powder Coating is a privately managed industrial unit founded and led by Mr. Arup Bera, located in Lakshmanpur Industrial Estate, Domjur, Howrah, West Bengal. Established with the vision of providing durable and high-quality surface finishing solutions, the company has steadily grown into a trusted name among local manufacturers and industrial suppliers. It primarily serves clients from the electrical, fabrication, furniture, and construction sectors who require corrosion-resistant, aesthetic, and long-lasting coatings for metallic products.

The company's operations cover a diverse range of products, including aluminium frames, battery boxes, beams, metal racks, steel furniture, almirahs, panel boxes, and driver desks. Balaji Powder Coating employs a small yet skilled workforce, specializing in pre-treatment, powder application, curing, and quality inspection. The organization maintains a flat structure that ensures direct supervision and hands-on management by Mr. Bera.

Balaji Powder Coating's mission is to deliver precision, durability, and consistency in every coated product while minimizing waste and turnaround time. Its vision is to evolve into a digitally optimized, high-efficiency coating facility capable of meeting growing industrial demands with reliability and sustainability.

Problem Statement

After multiple discussions with Mr. Arup Bera, staff members, and a detailed assessment of the production and storage areas, Balaji Powder Coating was identified to be facing challenges in the following areas:

Inventory Management:

- Water clogging (during rain) in the storage zone disrupts material handling and damages coated items awaiting dispatch.
- Lack of systematic storage layout and real-time tracking causes delays in locating items and managing order flow.
- Implementing a structured, data-backed inventory system and improving physical storage conditions are essential for efficiency and safety.

Processing Delays:

- Occur due to unorganized task sequencing and limited coordination between coating, curing, and dispatch sections.
- Lead to extended turnaround times and uneven workload distribution.
- Standardized workflow practices and scheduling optimization are required to ensure smoother operations.

Background of the Problem

Balaji Powder Coating operates in a highly competitive industrial environment where efficiency, timely delivery, and quality assurance are critical for customer satisfaction. The company manages a diverse range of products — including aluminium frames, battery boxes, beams, metal racks, steel furniture, almirahs, panel boxes, and driver desks — which require careful coordination between coating, curing, and dispatch operations. However, the current workflow and inventory management system are largely manual, leading to periodic inefficiencies and production delays.

One of the major challenges identified during discussions and on-site observations is water clogging in the storage and handling areas. This recurring issue not only disrupts the movement of goods but also affects the condition of coated materials awaiting dispatch, occasionally causing rework or loss. Additionally, the absence of a structured tracking mechanism makes it difficult to locate items quickly, monitor production status, or predict order readiness. These gaps contribute to extended turnaround times and fluctuating productivity levels.

Addressing these challenges is crucial for sustaining operational consistency and customer trust. By implementing a data-driven inventory management system alongside improved layout design and preventive measures for water clogging, Balaji Powder Coating can significantly reduce handling delays, protect product quality, and enhance operational

resilience. Streamlining these processes will ultimately enable the company to maintain reliability, improve delivery performance, and support long-term business growth under Mr. Arup Bera's leadership.

Problem Solving Approach

Balaji Powder Coating faces key operational challenges in inventory management, water clogging disturbances, and workflow inefficiencies during high-demand periods. To address these issues, a structured problem-solving framework combining data-driven analysis, systematic data collection, and MS Excel–based tools is proposed.

Key Components of the Problem-Solving Approach

Pareto Analysis of Customers:

Pareto Analysis will help Balaji identify the key customers contributing most to overall revenue. By applying the 80/20 principle, the company can determine which 20% of customers—such as **Unitech Projects** and **Steelwyn**—generate around 80% of the total income. This understanding enables Balaji to focus more on maintaining strong relationships with these major clients, ensuring timely order fulfillment, and providing customized service. Prioritizing high-revenue customers allows better allocation of resources, improved satisfaction levels, and stronger financial stability while still addressing the needs of smaller clients to ensure balanced business growth.

Trend Analysis for Order and Delivery Management:

Trend analysis of sales and delivery data will help understand demand fluctuations and identify peak periods. Historical data will be used to anticipate client requirements and prepare adequate stock without overburdening storage capacity. This proactive approach minimizes risks of overstocking or shortages and enhances on-time delivery performance even during heavy workflow periods.

Inventory Turnover Rate for Efficiency Evaluation:

The Inventory Turnover Rate will be calculated to assess how frequently stock is replenished and sold. Monitoring this metric helps detect stagnation due to space issues or environmental disturbances such as water clogging. Maintaining an optimal turnover rate ensures a healthy balance between supply and demand, better warehouse utilization, and improved cash flow management.

Intended Data Collection

Inventory Records:

Data on item names, serial numbers, prices, order dates, and delivery dates (as shown in the dataset) will be collected to track stock flow and value.

Sales Transactions:

Order frequency and client details will be monitored to evaluate which customers drive consistent demand and influence inventory planning.

Analysis Tools Using MS Excel

Pareto Analysis for Inventory Value:

Excel's sorting and charting tools will identify top-value inventory items, supporting better resource allocation and storage planning.

Trend Analysis for Demand Patterns:

Using Excel Pivot Tables and charts, sales and delivery timelines will be visualized to detect seasonality and customer behavior trends.

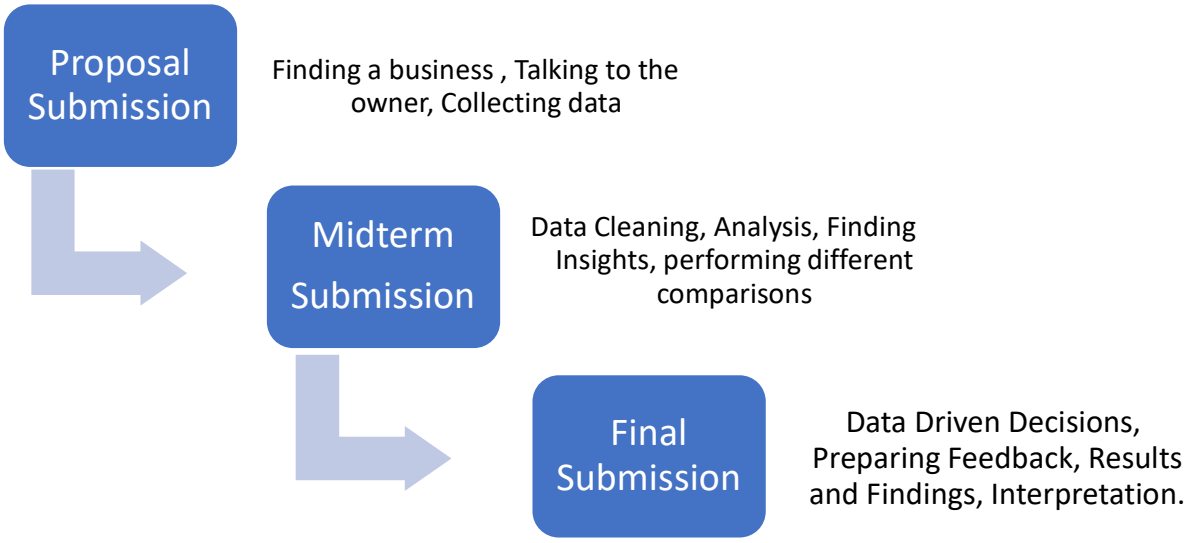
Inventory Turnover Calculation:

Formulas in Excel will compute turnover ratios, guiding decisions on restocking and space optimization.

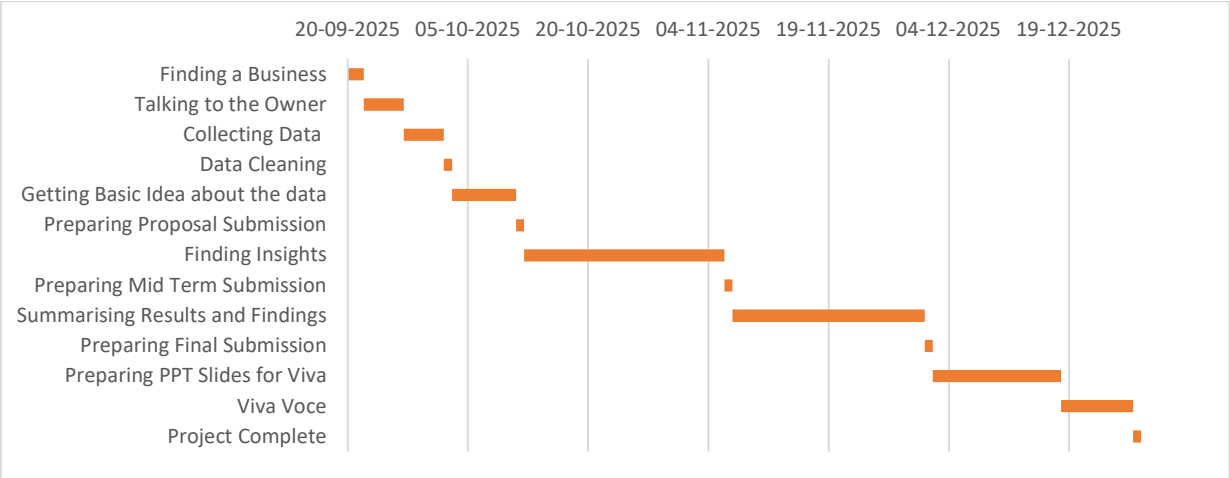
This structured, Excel-driven approach enables Balaji Powder Coating to strengthen inventory management, minimize disruptions, and enhance overall operational efficiency.

Expected TimeLine

I plan to collect data for at least 3months to justify the business problem. My main focus will be to gain insights into the business problem and provide suggestions. I will then monitor how the suggested changes impact the business over the next few months.



Gantt Chart



Expected Outcomes

The Problem-Solving Approach for **Balaji Powder Coating** is a strategic initiative designed to resolve key operational challenges and enhance the overall efficiency of its industrial processes. By leveraging data-driven techniques, the company aims to achieve measurable improvements in productivity, resource utilization, and customer satisfaction.

Enhanced Inventory Management:

Through Pareto Analysis and inventory turnover monitoring, Balaji Powder Coating will identify the most critical items contributing to inventory value. This will help reduce overstocking, prevent water-clogging disruptions, and ensure better control over raw materials and finished goods. Efficient inventory management will improve cash flow and minimize wastage.

Data-Driven Decision-Making:

Trend analysis of orders and production cycles will provide valuable insights into demand fluctuations and seasonal patterns. This data-centric approach will support better production planning, timely procurement, and streamlined workflow operations.

Operational Efficiency and Safety:

By optimizing layout and storage systems, the company will enhance workflow efficiency and minimize manual handling errors. Preventive measures for environmental issues such as water stagnation will further safeguard product quality.

Sustainable Growth:

Ultimately, this initiative will establish a foundation for sustainable growth—empowering Balaji Powder Coating to operate with greater accuracy, adaptability, and long-term competitiveness in the industrial finishing market.